

PoxHostAtlas: A reproducible cross-study atlas of host gene-expression remodeling during poxvirus infection

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Abstract

Poxviruses, including vaccinia virus, globally reprogram host gene expression upon infection. Public datasets for poxviruses extend through many viruses, hosts, cell types, and labs. It remains unknown whether host genes are altered reproducibly across independent studies rather than in single experiments. Here we present PoxHostAtlas, a curated collection of public poxvirus host-response RNA-sequencing datasets, reanalysed using a common bioinformatics workflow. Our analysis includes: PyDESeq2 differential expression, genome-wide DerSimonian-Laird random-effects meta-analysis with leave-one-study-out validation, host-program over-representation testing, cross-study co-expression network construction to rank host hubs, leave-one-dataset-out portability testing, and independent external validation.

Across the pooled datasets, 1,834 host genes were reproducibly changed (871 up, 963 down; ≥ 3 studies, consistent directionality, meta-FDR < 0.05 , $|\log_2FC| > 0.5$). Prominent programs were induction of inflammatory and NF- κ B signalling and down-regulation of cell cycle and ribosomal protein genes, consistent with host shutoff. Restricting analysis to vaccinia-only studies confirmed shutoff as the cleanest infection signature.

Within these genes, 29 form a small RNA-helicase module. Of these 29, DHX15 is the most consistently reproducible (meta-FDR $3.6e-19$) compared to DHX29 which remains reproducible only under specific contexts. A family-agnostic co-expression network identifies immune regulators (e.g., IL6, PTPRC, MAFB) as the most central host hubs. This signature also

transferred to held-out studies. An independent single-cell RNA-seq and proteomic dataset confirmed host shutoff across two cell types and at protein level.

PoxHostAtlas is a reusable and falsifiable resource to prioritize host genes for experimental follow-up. As a computational, hypothesis-generating resource, it does not make any causal claims.

Keywords: poxvirus, vaccinia, host shutoff, meta-analysis, reproducibility

Introduction

Vaccinia virus, the type species for poxviruses, is the best-characterized member of a large family encompassing the causative agents of smallpox and mpox. One of the defining features of poxvirus infection is the induction of host shutoff, in which the virus fully shuts off host transcription and translation while continuing translation of viral mRNAs. Park et al. identified RACK1- and eIF3-centered non-canonical translation initiation during poxvirus-induced shutoff [1]. In addition, host DEAD/DEAH-box RNA helicases such as DHX29 have been shown to regulate the tropism of the oncolytic myxoma virus [2], and DHX29 has been shown to cooperate with eIF3 to facilitate scanning of structured mRNAs [3]. However, there have been few studies looking at the relationship between poxviruses and host RNA helicases.

Although several dedicated poxvirus-host transcriptome studies have been conducted, they can be divided into two groups: (1) larger-scale, virus-centric, long-read sequencing efforts that characterize the transcriptome [4,5]; and (2) a smaller number of studies examining the host response to infection, including bulk RNA-seq together with ribosome profiling [6] and single-cell RNA-seq combined with proteomics [7].

Because these host datasets are from separate labs, using different viruses and cell types, an integrated analysis has never been conducted to determine which host genes and pathways are reproducibly altered across multiple independent datasets; if a gene shows a consistent pattern of change across multiple independent datasets, it is less likely to be a batch effect and more likely to represent true biological change. Here, we sought to answer an unbiased, whole-genome question: what host genes and pathways show reproducible alteration across independent datasets of poxvirus infection, and what are the most commonly targeted host hubs? We did not

limit our search to a particular protein family, because doing so would bias our findings, but instead we focused on the entire genome; we consider host translation and RNA helicase function as a pre-defined subset of the overall question.

Our resulting resource, PoxHostAtlas combines several public transcriptomic datasets to identify consistently targeted host genes, enabling their prioritization for experimental follow-up; the data were taken directly from publicly available datasets and re-analyzed here under one common workflow without re-claiming the findings of the original studies, making it a discovery tool rather than a definitive statement on mechanism. Because infection occurs over a relatively short window of time, one of our Tier-A datasets includes a time-course experiment, providing the opportunity to assess whether reproducible host alterations persist throughout the duration of infection. Inclusion of two distinct poxviruses also enables us to determine which host alterations occur across multiple different strains of poxviruses versus those that are specific to a single strain.

This reproducibility-first focus was a deliberate methodological decision: rather than highlighting what changed in any given experiment, we report only what changed in multiple independent experiments. Risks of batch effects and other analytical artifacts that could appear as real biological signals in a single dataset are therefore mitigated.

Materials and Methods

Data discovery and stratification

We searched publicly available repositories (GEO, SRA, ENA) and article supplementary materials for poxvirus host-transcriptome datasets according to pre-defined selection criteria and categorized by quality (Table 1). Datasets that met all criteria (Tier A) and contained count data

were included in the quantitative analyses. One additional dataset (Tier B) was selected for use as an independent validation set. Datasets that did not conform to the selection criteria and could not be compared to the other datasets (e.g., viral transcriptome datasets using long reads) were catalogued but not used in further analysis. Categorization prevents the inclusion of inappropriate data in the quantitative model. In cases where metadata inconsistencies were noted between the study text and the dataset file name, the file name was assumed to be correct and the discrepancy was recorded. An additional model was constructed where these samples were excluded to make sure that the main results were not affected.

Differential expression and meta-analysis

Count data was filtered for low counts, and differential expression was calculated between infected and control conditions using PyDESeq2 [8], an implementation of the DESeq2 model [9]. Identifiers for non-human (Vero) samples were mapped to human equivalents and collapsed to gene symbols using the GENCODE v44 database [10]. For each gene measured in ≥ 2 studies, a DerSimonian-Laird random-effects pooled log₂ fold change and corresponding heterogeneity measures (I^2 , τ^2) were calculated along with a directional concordance measure and Benjamini-Hochberg adjusted meta-FDR. Leave-one-out re-pooling was used to assess stability of each pooled estimate. Genes were considered consistently deregulated if they were measured in ≥ 3 studies, had consistent directionality in each dataset that measured them, were significantly deregulated (meta-FDR < 0.05), and changed by > 0.5 log₂ fold change.

Enrichment, network-based prioritization, and independent validation

The reproducibly deregulated genes were assessed for enrichment (hypergeometric test, Benjamini-Hochberg adjusted) for various host pathways (interferon-stimulated genes, inflammation, NF- κ B signaling, integrated stress response, cell cycle/mitosis, cholesterol biosynthesis, apoptosis, RNA helicases, translation initiation factors, ribosomal proteins) reported in the literature and defined as “programs”. A background of all genes measured in ≥ 3 studies was used and the most common direction of change was utilized as the test statistic. Host pathway directionality was also calculated from the 4 vaccinia contrasts (drawn from two studies) that contained vaccinia infection only, to distinguish general infection-related effects from effector-specific effects. Prioritization of consistently targeted host genes that were not identified by the program search was done by constructing a co-expression network across studies. The 45-sample gene expression matrix was centered within each study and pairwise correlations were calculated for every sample. Edges were included if the absolute Pearson correlation was ≥ 0.7 . Network centrality was calculated utilizing a combination of directed community detection and edge clustering on the subset of positive edges (STRING physical interaction ranking is being explored as a second ranking) [11]. Finally, multiple classification algorithms (elastic net logistic regression, random forest, gradient boosting, linear support vector machine) with bootstrap confidence intervals were evaluated via leave-one-dataset-out cross-validation to classify samples as infected or uninfected, with whole studies held out, and multiple controls (label permutation, random gene selection, ablation, identity of study) implemented. We additionally evaluated an independent single-cell RNA sequencing and proteomic dataset for vaccinia infection [7] to see if we can find a similar effect in other cell types and at the protein level.

Results

A curated public poxvirus atlas

The four Tier-A datasets that were included in the screening incorporate two poxviruses (Myxoma and Vaccinia), different human cells, and a time-course experiment with non-human primates, while the Tier-B validation dataset includes one study (refer to Table 1 and Figure 1). All these experiments resulted in the generation of 10 harmonized contrasts and quantification of 16,841 host genes at least from three experiments involving different viruses, cell types, and laboratories.

Figure 1. PoxHostAtlas: systematic public poxvirus host-response atlas, tiering, and pipeline

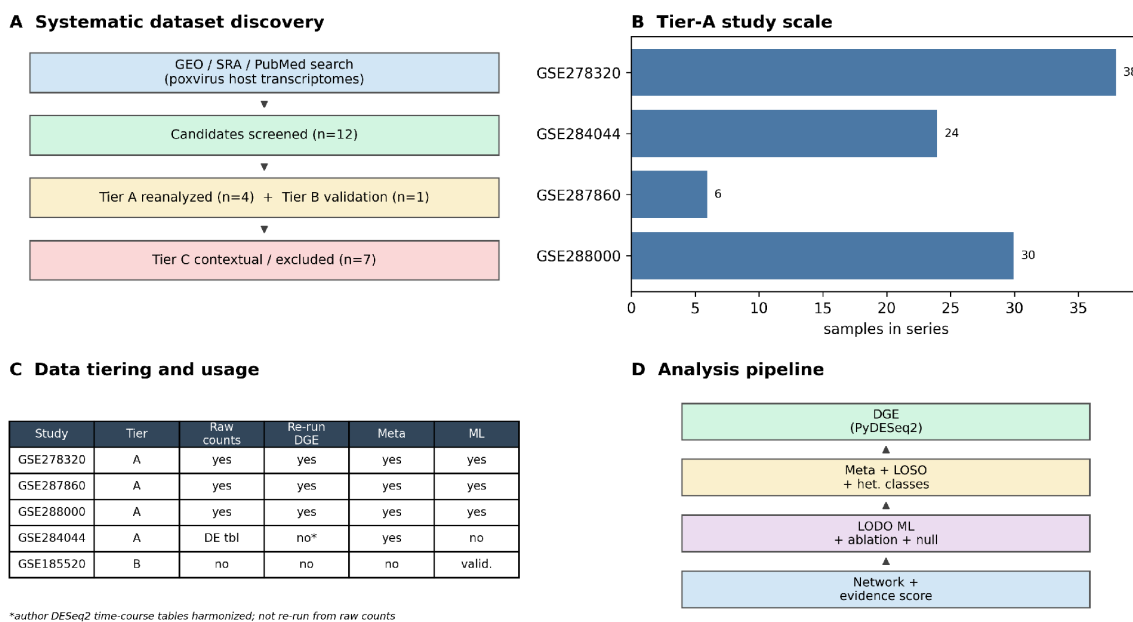


Figure 1. Systematic dataset discovery, scale, virus coverage and the end-to-end analysis pipeline.

Poxvirus infection reproducibly remodels a broad host program

A total of 1,834 host genes were consistently modified across the genome - 871 of these were up-regulated and 963 down-regulated (see Figure 2). Of these genes, only 29 genes (1.6%)

were either RNA helicases or translation factors, which shows that there was no single gene family dominating the signal. Rather the signal can be found in two programs, as seen in Figure 3: inflammatory and NF- κ B signaling (with genes such as IL6, IL1B, CXCL8, and BIRC3), with the former showing a significant increase (over-represented by a factor of 5.4, and with false discovery rate [FDR] equal to $3e-7$). Cell-cycle and mitotic genes (such as CDK1, CCNB1/CCNB2, PLK1, and FOXM1) showed considerable repression (a factor of 5.7, FDR = $3e-7$), together with systematic down-regulation of ribosomal-protein genes (24 out of 26 of them were down-regulated). Among all these genes, the most significantly induced ones were transcripts related to the immune and inflammation consortia (including IL6, IL1B, and BIRC3), while the most down-regulated genes were proliferation- and metabolism-related (e.g., DEPTOR and IGF1). This picture confirms that major changes are present indeed across the entire gene network.

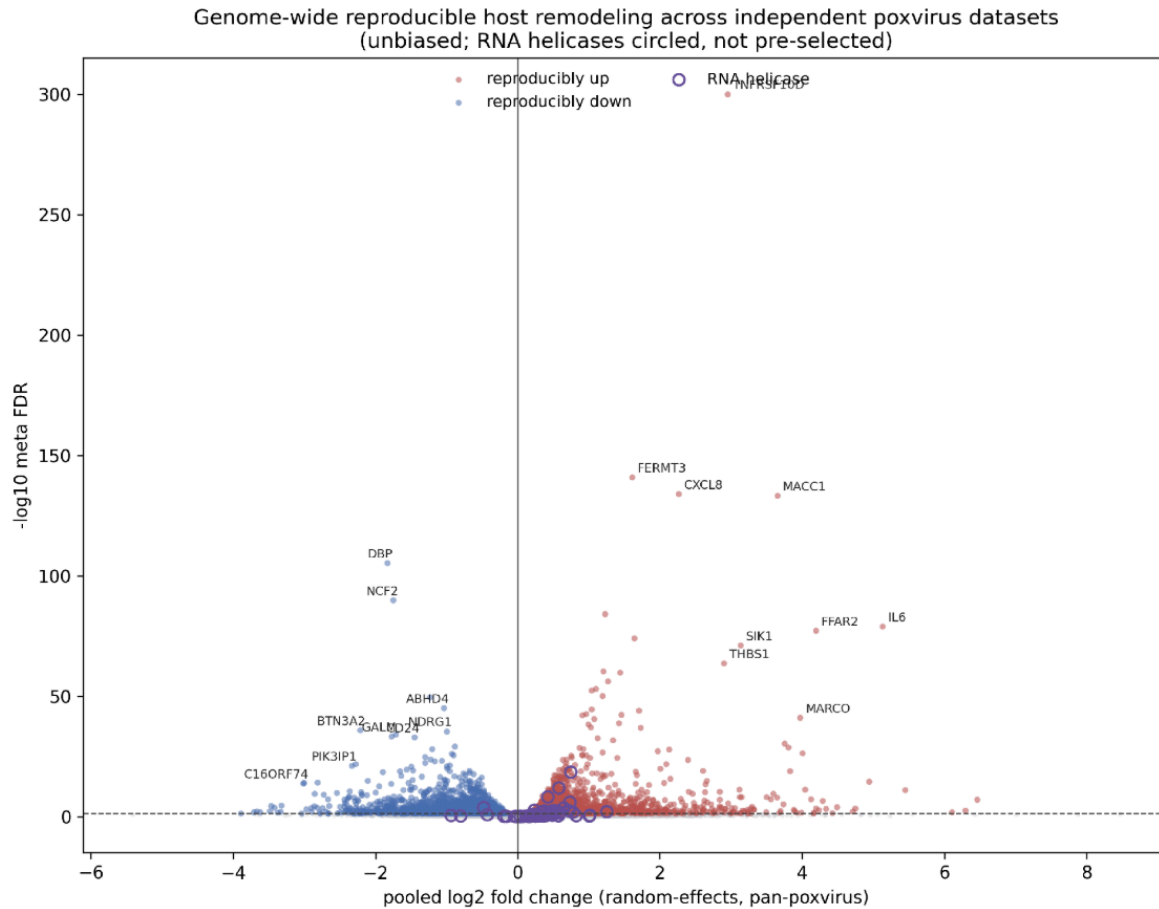


Figure 2. Genome-wide random-effects meta-analysis volcano; RNA helicases circled, not pre-selected.

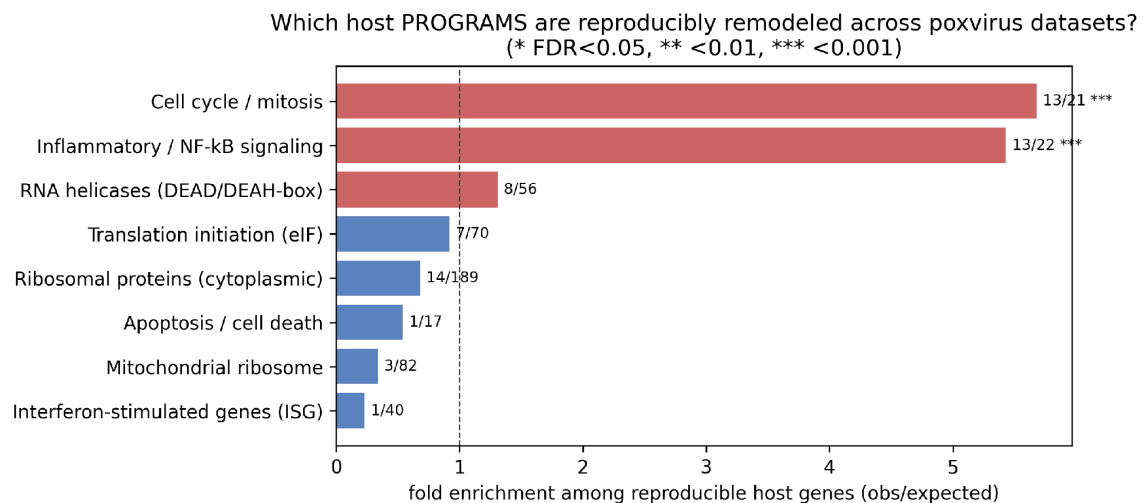


Figure 3. Host programs over-represented among reproducibly remodeled genes.

Since the two datasets pertain to Myxoma effector contrasts where their effectors control inflammation, we recalculated the program directions just for the vaccinia-infection studies. The clearest signature of infection in this case was host shutoff (with its ribosomal proteins median log2 fold-change close to -0.79, cell cycle about -0.58, and translation initiation factors equal to about -0.32), while the inflammatory induction, although present, was considerably lower (about +0.74 vs. +2.0 in the aggregate). Therefore, the cleanest and most consistent pan-poxvirus infection signature is host shutoff; the stronger inflammatory induction seen in the aggregate is driven partly by the Myxoma effector contrasts.

As one Tier-A dataset is a time-course study of vaccinia (2, 6, and 24 h post-infection), it was possible to analyze the replicable shutoff signature throughout the time course rather than in a single time point. The down-regulated cell cycle and ribosome programs were seen in all contrasts of the time course, not only in the initial time point, and this points toward a long-term rather than transient nature of the host response. This is assessed carefully, given that it is based solely on one study in the non-human-primate cell line and mapped to human homologues, but its agreement with human datasets indicates the reproducible programs' nature as a response caused by the infection rather than by a particular cell type or time point.

Network ranking and the helicase sub-program

A non-family-centered co-expression network (228 nodes and 4,430 edges) has successfully identified the presence of a co-up-regulatory module containing 128 genes and a co-down regulatory module comprising 93 genes. Additionally, it has positioned immune regulators such as IL6, PTPRC, and MAFB with the highest degree of replication in their main role as central hubs (Figure 4); the highest-ranked candidates are listed in Table 2. After

analyzing various responses, it is worth mentioning that 18 of the 19 significant DDX/DHX helicases were up-regulated, although the family as a whole was not over-represented among the reproducible genes (1.3-fold; not significant). Within the family, the lowest meta-FDR was that of DHX15 (3.6e-19; top 0.6% of all genes) and this was reproducible across studies; however DHX29 was only reproducible in specific contexts (I^2 about 80%; meta-FDR approximately 0.06).

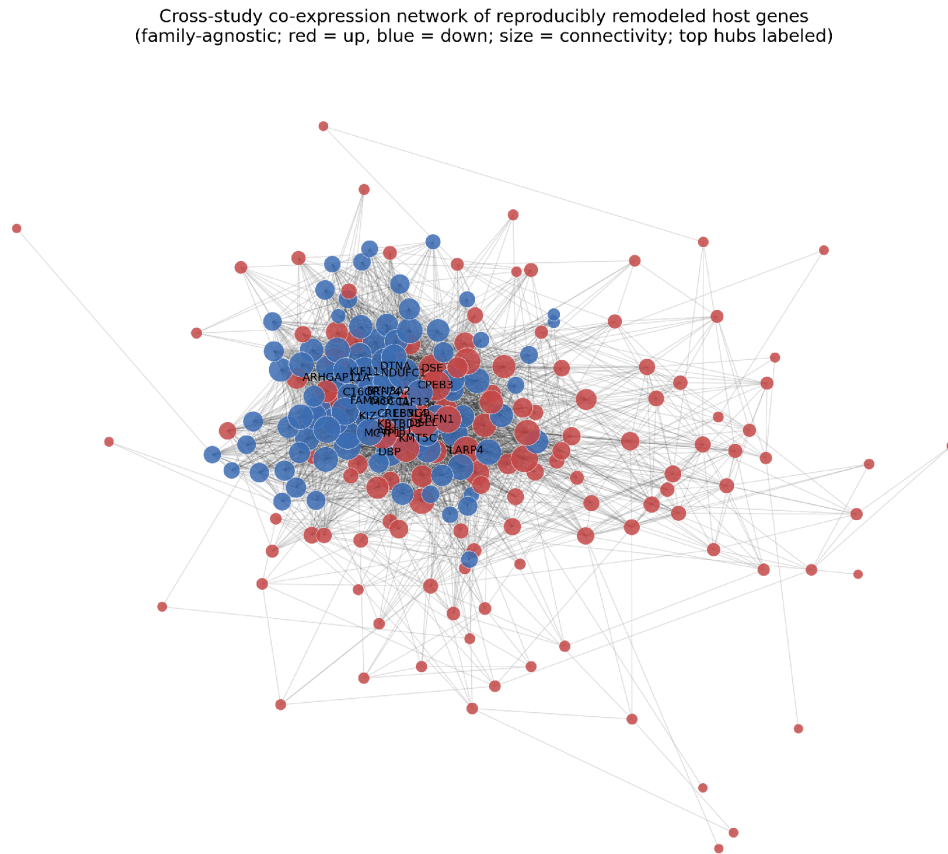


Figure 4. Cross-study co-expression network of reproducibly remodelled host genes.

Figure 2. Cross-study meta-analysis and heterogeneity classification

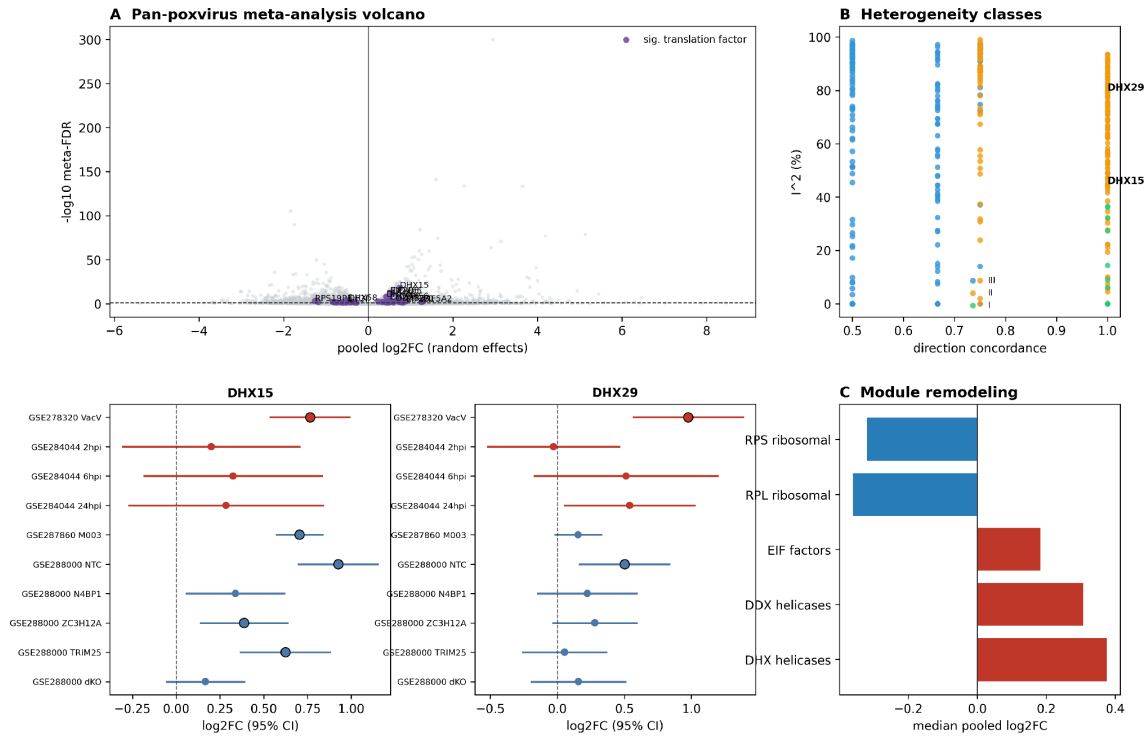


Figure 5. Translation/helicase sub-program: DHX15 (conserved) and DHX29 (context-dependent).

Reproducibility across studies, modalities and cell types

Using a leave-one-dataset-out classification model, the test results demonstrated that these models were effective at determining whether a sample was or was not infected (balanced accuracy 0.79). With randomly permuted labels, the performance fell to chance (0.54). On the other hand, the identity of the study was highly predictable itself (0.93), thus confirming the batch structure. With the top-ranked features removed, the model was degraded beyond the 99.8th percentile of random ablations ($p=0.003$).

Heterogeneity classification indicated that 33 of 485 genes fell in a conserved low-heterogeneity class, 254 conserved but high-heterogeneity, and 198 context-dependent (Table 3, Figure 6). Leave-one-study-out re-pooling showed a preserved effect direction for 67.4%; in all of the four re-poolings, DHX15 retained direction and significance.

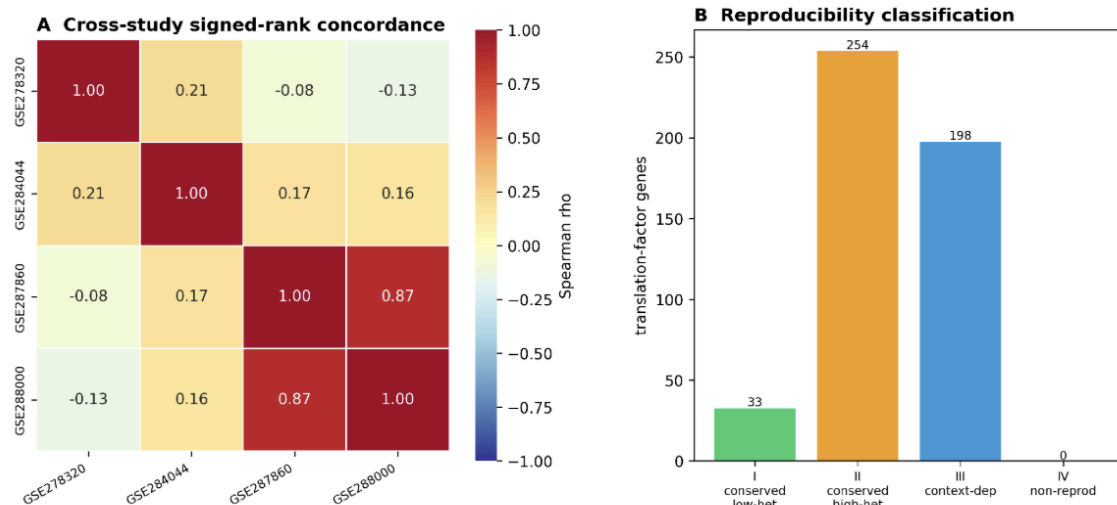


Figure 6. Cross-study rank concordance and gene reproducibility classes.

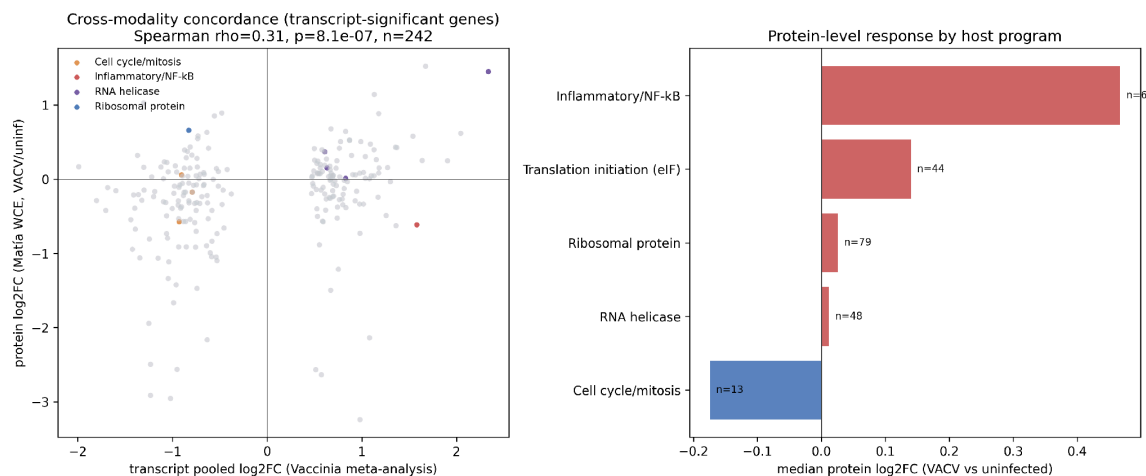


Figure 7. Independent external validation: transcript-level changes concordant with an independent proteome.

When all gathered evidence is analyzed, it becomes apparent that a single dataset was not responsible for this high degree of replicability observed in the studied genes. Remarkably the differences in gene regulation seem to be concentrated in genes within low-heterogeneity signaling pathways. Notably, the most significantly increased expression was identified for the following inflammatory genes: IL6 and IL1B. At the same time, it was discovered that the most evident downregulation in the expression was indicated for genes related to proliferation and metabolism.

Discussion

The poxvirus infection alters the programming of the host to induce immune response signals while inhibiting mechanisms that lead to cell replication [1]. Since the results of the experiments conducted in several laboratories show the overall effect of poxvirus irrespective of local environmental effects, the reproducible induction of immune signalling and proliferation suppression can be attributed to the infection itself and not to the local conditions. Gene expression data from several labs indicate that the inflammatory response from a poxvirus infection is consistently amplified and the virus does so while shutting off host translation and cell-cycle programs. Based on the considerations indicated above, several helicases, notably DHX15, can be classified as potential candidates warranting further research.

In biological terms, the reproducible program makes sense. The silencing of ribosome proteins and translation initiation genes as well as cell cycle arrest relates to the type of virus that takes over the host's translation machinery while causing the host cells to stop dividing so that the virus can replicate. During such events, the reproducible expression of genes linked to inflammation and interferon-related processes takes place. The fact that both of these elements can be reproduced using different viruses, hosts, and labs means that they are common elements of poxvirus infection, which is particularly important because orthopoxviruses like mpox pose a growing public health threat today.

Limitations

All primary signals are transcript-level, with one independent proteomic validation; for mechanistic claims, ribosome profiling and perturbation experiments are needed. The pooled

meta-analysis confounds true infection with Myxoma effector contrasts, hence the caveat about the vaccinia-only sensitivity layer; because there are only two studies that are Tier-A and vaccinia, that layer cannot apply the criterion for ≥ 3 -study and is reported as a directional sensitivity analysis. Cross-species orthologue mapping adds another potential confounder, as does public metadata noise; the co-expression network is a regulon, not necessarily interactions. The study is computational/hypothesis-generating, and does not claim primary discovery or causal function.

Conclusion

PoxHostAtlas is a reusable resource whose results are capable of being independently verified. Infection of poxvirus remodels reproducibly a broad host program consisting of immune induction along with cell-cycle and translation machinery suppression. These programs persist regardless of the virus, the host organism, or even the research laboratory and therefore are unlikely to be batch artifacts. We keep individual claims honest by using a genome-wide approach, rather than family-first; within that program, DHX15 emerges as the helicase candidate that is the most reproducible for experimental testing. With more availability of host-response datasets, the same openly released pipeline can incorporate them and refine these rankings.

Data and Code Availability

The entire analytical framework, from the analysis scripts and database, methods for data retrieval, differential analysis and meta-analysis results, and analyses (including figures), can be found in the project's repository along with the SHA-256 file for integrity authentication. The

raw data can be accessed using public identifiers (GEO GSE278320, GSE284044, GSE287860, GSE288000 and GSE185520).

Table 1. Curated poxvirus datasets and their role in the analysis.

Dataset	Tier	Virus / host	n samples	Role
GSE278320	A	Vaccinia / human HAP1	38	Anchor; meta-analysis; stress-test
GSE284044	A	Vaccinia / Vero (time course)	24	Meta-analysis (temporal)
GSE287860	A	Myxoma effector / human	6	Meta-analysis; stress-test
GSE288000	A	Myxoma effector / human (KO panel)	30	Meta-analysis; stress-test
GSE185520	B	Vaccinia D10 / human	30	External directional validation
Matia 2024	External	Vaccinia / human (single-cell + proteomics)	—	Independent cross-modality validation

Table 2. Top-ranked reproducible host hubs prioritized for experimental follow-up.

Gene	Program	Direction	Pooled log2FC	meta-FDR	Degree	Hub score
DTNA	other	up	+1.64	8.4e-75	89	0.956
DBP	other	down	-1.84	5.4e-106	82	0.952
BTN3A2	other	down	-2.22	1.4e-36	99	0.941
LARP4	other	up	+1.04	2.2e-45	82	0.921
DSE	other	up	+1.34	2.2e-32	84	0.904
ABHD4	other	down	-1.04	8.0e-46	74	0.893
IL6	Inflammatory/N F-kB	up	+5.13	1.2e-79	68	0.884
GALM	other	down	-1.77	5.4e-34	73	0.875
SATB1	other	down	-0.90	2.8e-26	79	0.871
MACC1	other	up	+3.65	6.1e-134	62	0.866
ABTB1	other	down	-1.50	1.9e-21	96	0.864
LENG9	other	up	+2.00	1.1e-20	95	0.860

Ranked by network hub score among reproducibly remodelled genes. Full ranking is available in the project repository.

Table 3. Reproducibility classification and leave-one-study-out stability.

Class	Definition	Genes
I — conserved, low heterogeneity	Concordant direction, I^2 low	33
II — conserved, high heterogeneity	Concordant direction, I^2 high	254
III — context-dependent	Direction varies by context	198
Total classified		485
Direction preserved in every leave-one-study-out re-pooling		327 (67.4%)
DHX15	Direction and significance preserved in all four re-poolings	✓

Supporting Information

Table S1. Complete dataset screening record.

Table S2. Host-program over-representation, direction-stratified.

Table S3. Program direction: pooled versus vaccinia-only.

Table S4. Reproducibility classification and leave-one-study-out stability.

Table S5. Top 15 reproducibly up- and down-regulated host genes.

Table S6. Machine-learning controls.

Table S7. Key analysis parameters.

Figures S1–S17. Pipeline, full-size helicasе forests, vaccinia-only volcano, temporal dynamics, host-background robustness, cross-study rank correlation, ranked hubs, STRING network, evidence matrices, single-cell and cross-modality validation, scope statement.

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